Course Title: Database Management Systems

Course Code: UE24CS351A

Academic Session: Aug-Dec 2026

DBMS Experiential Learning – Level 3 Project

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Title : Defence Asset Readiness and Maintanance Dependency System

PART 1: INSERT DATA :

```
mysql> INSERT INTO COUNTRY
    -> (CountryID, CountryName, Region)
    -> VALUES
    -> (1, 'India', 'South Asia'),
    -> (2, 'France', 'Europe'),
    -> (3, 'Japan', 'East Asia');
Query OK, 3 rows affected (0.32 sec)
Records: 3  Duplicates: 0  Warnings: 0

mysql> INSERT INTO `USER`
    -> (UserID, UserName, PasswordHash, FullName, Role, Email, Phone)
    -> VALUES
    -> (101, 'analyst01', 'hash_demo_101', 'Arjun Rao',
    ->  'Analyst', 'analyst01@example.com', '9000000001'),
    -> (102, 'admin01', 'hash_demo_102', 'Kiran Sharma',
    ->  'Admin', 'admin01@example.com', '9000000002'),
    -> (103, 'inspector01', 'hash_demo_103', 'Vikram Nair',
    ->  'Inspector', 'inspector01@example.com', '9000000003');
Query OK, 3 rows affected (0.04 sec)
Records: 3  Duplicates: 0  Warnings: 0

mysql> INSERT INTO SERVICE
    -> (ServiceID, ServiceName, Description, CountryID)
    -> VALUES
    -> (1, 'Indian Air Force', 'Air service of India', 1),
    -> (2, 'Indian Navy', 'Naval service of India', 1),
    -> (3, 'Indian Army', 'Land service of India', 1);
Query OK, 3 rows affected (0.03 sec)
Records: 3  Duplicates: 0  Warnings: 0
```

```
mysql> INSERT INTO DEFENCE_INDICATOR
    -> (IndicatorID, CountryID, Year, DefenceBudget,
    ->  RevenueExpenditure, CapitalExp, MaintenanceExp,
    ->  IndigenisationIndex, Notes)
    -> VALUES
    -> (1001, 1, 2024, 620000.00, 360000.00, 260000.00,
    ->  85000.00, 68.50, 'Synthetic demonstration data'),
    -> (1002, 1, 2025, 680000.00, 380000.00, 300000.00,
    ->  92000.00, 72.00, 'Synthetic demonstration data'),
    -> (1003, 2, 2025, 610000.00, 350000.00, 260000.00,
    ->  88000.00, 70.00, 'Synthetic demonstration data'),
    -> (1004, 3, 2025, 540000.00, 330000.00, 210000.00,
    ->  76000.00, 65.00, 'Synthetic demonstration data');
Query OK, 4 rows affected (0.03 sec)
Records: 4  Duplicates: 0  Warnings: 0
```

```
mysql> INSERT INTO UNIT
    -> (UnitID, UnitName, Type, Location, ServiceID)
    -> VALUES
    -> (201, 'Fighter Squadron Alpha', 'Formation', 'Bengaluru', 1),
    -> (202, 'Transport Squadron Bravo', 'Formation', 'Chandigarh', 1),
    -> (203, 'Naval Support Unit', 'Base', 'Kochi', 2),
    -> (204, 'Armoured Unit Charlie', 'Formation', 'Jaisalmer', 3);
Query OK, 4 rows affected (0.04 sec)
Records: 4  Duplicates: 0  Warnings: 0

mysql> INSERT INTO ASSET_TYPE
    -> (AssetTypeID, AssetTypeName, Category, Description)
    -> VALUES
    -> (301, 'Fighter Aircraft', 'Aircraft',
    ->  'Synthetic fighter aircraft asset category'),
    -> (302, 'Transport Aircraft', 'Aircraft',
    ->  'Synthetic transport aircraft category'),
    -> (303, 'Naval Support Platform', 'Naval',
    ->  'Synthetic naval support asset category'),
    -> (304, 'Armoured Vehicle', 'Land',
    ->  'Synthetic armoured vehicle category');
Query OK, 4 rows affected (0.04 sec)
Records: 4  Duplicates: 0  Warnings: 0

mysql> INSERT INTO ASSET
    -> (AssetID, AssetName, IdentificationNo, Model,
    ->  MnfYear, Status, CriticalityLevel, UnitID, AssetTypeID)
    -> VALUES
    -> (401, 'Fighter Asset A-101', 'IAF-SYN-101',
    ->  'Fighter-Mk1', 2023, 'Ready', 'High', 201, 301),
    -> (402, 'Fighter Asset A-102', 'IAF-SYN-102',
    ->  'Fighter-Mk1', 2022, 'Degraded', 'High', 201, 301),
    -> (403, 'Transport Asset T-201', 'IAF-SYN-201',
    ->  'Transport-X', 2021, 'Ready', 'Medium', 202, 302),
    -> (404, 'Naval Asset N-301', 'IN-SYN-301',
    ->  'Naval-X', 2024, 'Ready', 'High', 203, 303),
    -> (405, 'Armoured Asset L-401', 'IA-SYN-401',
    ->  'Armour-X', 2022, 'Unavailable', 'High', 204, 304);
Query OK, 5 rows affected (0.03 sec)
Records: 5  Duplicates: 0  Warnings: 0
```

```
mysql> INSERT INTO FAILURE_EVENT
    -> (FailureID, FailureType, Severity, OccurredAt,
    ->  Description, AssetID)
    -> VALUES
    -> (501, 'Cooling System Failure', 'High',
    ->  '2026-09-20 10:30:00',
    ->  'Abnormal cooling performance detected', 402),
    -> (502, 'Hydraulic Failure', 'Medium',
    ->  '2026-09-21 14:15:00',
    ->  'Hydraulic pressure below expected range', 405),
    -> (503, 'Power System Failure', 'High',
    ->  '2026-09-23 09:45:00',
    ->  'Power subsystem malfunction detected', 401);
Query OK, 3 rows affected (0.03 sec)
Records: 3  Duplicates: 0  Warnings: 0
```

```
mysql> INSERT INTO INSPECTION
    -> (InspectionID, FailureID, InspectorID,
    ->  InspectionType, InspectionDate, Result, Remarks)
    -> VALUES
    -> (601, 501, 103,
    ->  'Technical Inspection', '2026-09-20',
    ->  'Fail', 'Cooling component requires replacement'),
    -> (602, 502, 103,
    ->  'Technical Inspection', '2026-09-21',
    ->  'Pass', 'Temporary issue identified'),
    -> (603, 503, 103,
    ->  'Technical Inspection', '2026-09-23',
    ->  'Fail', 'Power component requires maintenance');
Query OK, 3 rows affected (0.03 sec)
Records: 3  Duplicates: 0  Warnings: 0

mysql> INSERT INTO MAINTENANCE_WORK_ORDER
    -> (WorkOrderID, AssetID, Description,
    ->  OrderDate, Priority, Status)
    -> VALUES
    -> (701, 402,
    ->  'Repair cooling system',
    ->  '2026-09-20', 'High', 'InProgress'),
    -> (702, 405,
    ->  'Inspect hydraulic system',
    ->  '2026-09-21', 'Medium', 'Closed'),
    -> (703, 401,
    ->  'Repair power subsystem',
    ->  '2026-09-23', 'High', 'Open');
Query OK, 3 rows affected (0.03 sec)
Records: 3  Duplicates: 0  Warnings: 0

mysql> INSERT INTO MAINTENANCE_EVENT
    -> (MaintenanceID, WorkOrderID, Remarks,
    ->  PerformedBy, StartDateTime, EndDateTime,
    ->  MaintenanceType, ActionTaken,
    ->  DowntimeHours, CostIncurred)
    -> VALUES
    -> (801, 702,'Hydraulic inspection completed', 103, '2026-09-21 16:00:00', '2026-09-21 18:00:00',
    ->  'Inspection','Hydraulic system inspected',2.00,25000.00),
    -> (802, 701,'Cooling system repair initiated',103,'2026-09-22 09:00:00','2026-09-22 15:30:00',
    ->  'Repair','Cooling component replaced',6.50,85000.00);
Query OK, 2 rows affected (0.03 sec)
Records: 2  Duplicates: 0  Warnings: 0
```

```
mysql> INSERT INTO SUBSYSTEM
    -> (SubsystemID, SubsystemName, Description,
    ->  CriticalityLevel, AssetID)
    -> VALUES
    -> (901, 'Engine System',
    ->  'Primary propulsion subsystem',
    ->  'High', 401),
    -> (902, 'Cooling System',
    ->  'Thermal management subsystem',
    ->  'High', 402),
    -> (903, 'Hydraulic System',
    ->  'Hydraulic control subsystem',
    ->  'High', 405),
    -> (904, 'Power System',
    ->  'Electrical power subsystem',
    ->  'High', 401);
Query OK, 4 rows affected (0.03 sec)
Records: 4  Duplicates: 0  Warnings: 0
```

```
mysql> INSERT INTO COMPONENT
    -> (ComponentID, ComponentName, Description,
    ->  CriticalityLevel, IsRepairable, SubsystemID)
    -> VALUES
    -> (1001, 'Fuel Pump',
    ->  'Synthetic fuel delivery component',
    ->  'High', 'Yes', 901),
    -> (1002, 'Engine Control Unit',
    ->  'Synthetic engine control component',
    ->  'High', 'Yes', 901),
    -> (1003, 'Cooling Pump',
    ->  'Synthetic cooling component',
    ->  'High', 'Yes', 902),
    -> (1004, 'Hydraulic Valve',
    ->  'Synthetic hydraulic control component',
    ->  'High', 'Yes', 903),
    -> (1005, 'Power Controller',
    ->  'Synthetic electrical control component',
    ->  'High', 'Yes', 904);
Query OK, 5 rows affected (0.03 sec)
Records: 5  Duplicates: 0  Warnings: 0

mysql> INSERT INTO COMPONENT_DEPENDENCY
    -> (ComponentID, DependsOnComponentID)
    -> VALUES
    -> (1002, 1001),
    -> (1003, 1001),
    -> (1005, 1002);
Query OK, 3 rows affected (0.03 sec)
Records: 3  Duplicates: 0  Warnings: 0
```

```
mysql> INSERT INTO SPARE_PART
    -> (SparePartID, PartName, PartNumber,
    ->  UOM, StockQuantity, ReorderLevel,
    ->  UnitCost, CriticalityLevel)
    -> VALUES
    -> (1101, 'Fuel Pump Seal', 'FPS-001',
    ->  'Piece', 50, 10, 5000.00, 'High'),
    -> (1102, 'Cooling Pump Assembly', 'CPA-002',
    ->  'Piece', 8, 3, 85000.00, 'High'),
    -> (1103, 'Hydraulic Valve Kit', 'HVK-003',
    ->  'Kit', 15, 5, 25000.00, 'Medium'),
    -> (1104, 'Power Controller Module', 'PCM-004',
    ->  'Piece', 6, 2, 120000.00, 'High');
Query OK, 4 rows affected (0.03 sec)
Records: 4  Duplicates: 0  Warnings: 0

mysql> INSERT INTO COMPONENT_SPARE
    -> (ComponentID, SparePartID, QuantityRequired, IsCritical)
    -> VALUES
    -> (1001, 1101, 2, 'Yes'),
    -> (1003, 1102, 1, 'Yes'),
    -> (1004, 1103, 1, 'Yes'),
    -> (1005, 1104, 1, 'Yes');
Query OK, 4 rows affected (0.03 sec)
Records: 4  Duplicates: 0  Warnings: 0
```

```
mysql> INSERT INTO SUPPLIER
    -> (SupplierID, SupplierName, ContactPerson,
    ->  Phone, Email, Address)
    -> VALUES
    -> (1201, 'Bharat Defence Components',
    ->  'Rohan Mehta',
    ->  '9000012001',
    ->  'supplier1@example.com',
    ->  'Bengaluru'),
    -> (1202, 'Aero Systems Manufacturing',
    ->  'Neha Rao',
    ->  '9000012002',
    ->  'supplier2@example.com',
    ->  'Hyderabad'),
    -> (1203, 'Precision Engineering Works',
    ->  'Amit Kumar',
    ->  '9000012003',
    ->  'supplier3@example.com',
    ->  'Pune');
Query OK, 3 rows affected (0.03 sec)
Records: 3  Duplicates: 0  Warnings: 0

mysql> INSERT INTO SUPPLIER_SPARE
    -> (SupplierID, SparePartID, LeadTimeDay,
    ->  UnitPrice, PreferredSupplier)
    -> VALUES
    -> (1201, 1101, 7, 5000.00, 'Yes'),
    -> (1202, 1102, 14, 85000.00, 'Yes'),
    -> (1203, 1103, 10, 25000.00, 'No'),
    -> (1202, 1104, 21, 120000.00, 'Yes');
Query OK, 4 rows affected (0.03 sec)
Records: 4  Duplicates: 0  Warnings: 0
```

PART 2: UPDATE

1. Update asset status

```
mysql> select * from asset where AssetID = 401;
+---------+--------------------+-----------------+------------+---------+--------+-----------------+--------+-------------+
| AssetID | AssetName          | IdentificationNo | Model      | MnfYear | Status | CriticalityLevel | UnitID | AssetTypeID |
+---------+--------------------+-----------------+------------+---------+--------+-----------------+--------+-------------+
|     401 | Fighter Asset A-101 | IAF-SYN-101     | Fighter-Mk1 |    2023 | Ready  | High            |    201 |         301 |
+---------+--------------------+-----------------+------------+---------+--------+-----------------+--------+-------------+
1 row in set (0.01 sec)

mysql> UPDATE ASSET SET Status = 'Degraded' WHERE AssetID = 401;
Query OK, 1 row affected (0.01 sec)
Rows matched: 1  Changed: 1  Warnings: 0

mysql> select * from asset where AssetID = 401;
+---------+--------------------+-----------------+------------+---------+----------+-----------------+--------+-------------+
| AssetID | AssetName          | IdentificationNo | Model      | MnfYear | Status   | CriticalityLevel | UnitID | AssetTypeID |
+---------+--------------------+-----------------+------------+---------+----------+-----------------+--------+-------------+
|     401 | Fighter Asset A-101 | IAF-SYN-101     | Fighter-Mk1 |    2023 | Degraded | High            |    201 |         301 |
+---------+--------------------+-----------------+------------+---------+----------+-----------------+--------+-------------+
1 row in set (0.00 sec)
```

2. Update maintenance work order

```
mysql> select * from MAINTENANCE_WORK_ORDER where WorkOrderID = 701;
+-------------+---------+-----------------------+------------+----------+------------+
| WorkOrderID | AssetID | Description           | OrderDate  | Priority | Status     |
+-------------+---------+-----------------------+------------+----------+------------+
|         701 |     402 | Repair cooling system | 2026-09-20 | High     | InProgress |
+-------------+---------+-----------------------+------------+----------+------------+
1 row in set (0.01 sec)

mysql> UPDATE MAINTENANCE_WORK_ORDER SET Status = 'Completed' WHERE WorkOrderID = 701;
Query OK, 1 row affected (0.01 sec)
Rows matched: 1  Changed: 1  Warnings: 0

mysql> select * from MAINTENANCE_WORK_ORDER where WorkOrderID = 701;
+-------------+---------+-----------------------+------------+----------+------------+
| WorkOrderID | AssetID | Description           | OrderDate  | Priority | Status     |
+-------------+---------+-----------------------+------------+----------+------------+
|         701 |     402 | Repair cooling system | 2026-09-20 | High     | Completed |
+-------------+---------+-----------------------+------------+----------+------------+
1 row in set (0.00 sec)
```

3. **Update spare stock**

```
mysql> select * from SUPPLIER_SPARE WHERE SupplierID = 1203 AND SparePartID = 1103;
+------------+-------------+-------------+-----------+-------------------+
| SupplierID | SparePartID | LeadTimeDay | UnitPrice | PreferredSupplier |
+------------+-------------+-------------+-----------+-------------------+
|       1203 |        1103 |          10 |  25000.00 | No                |
+------------+-------------+-------------+-----------+-------------------+
1 row in set (0.00 sec)

mysql> UPDATE SUPPLIER_SPARE SET PreferredSupplier = 'Yes' WHERE SupplierID = 1203 AND SparePartID = 1103;
Query OK, 1 row affected (0.01 sec)
Rows matched: 1  Changed: 1  Warnings: 0

mysql> select * from SUPPLIER_SPARE WHERE SupplierID = 1203 AND SparePartID = 1103;
+------------+-------------+-------------+-----------+-------------------+
| SupplierID | SparePartID | LeadTimeDay | UnitPrice | PreferredSupplier |
+------------+-------------+-------------+-----------+-------------------+
|       1203 |        1103 |          10 |  25000.00 | Yes               |
+------------+-------------+-------------+-----------+-------------------+
1 row in set (0.01 sec)
```

PART 3: DELETE

1. **Delete a dependency relationship.**

```
mysql> select ComponentID from COMPONENT_DEPENDENCY;
+-------------+
| ComponentID |
+-------------+
|        1002 |
|        1003 |
|        1005 |
+-------------+
3 rows in set (0.00 sec)

mysql> DELETE FROM COMPONENT_DEPENDENCY WHERE ComponentID = 1005 AND DependsOnComponentID = 1002;
Query OK, 1 row affected (0.01 sec)

mysql> select ComponentID from COMPONENT_DEPENDENCY;
+-------------+
| ComponentID |
+-------------+
|        1002 |
|        1003 |
+-------------+
2 rows in set (0.00 sec)
```

2. **Delete a supplier-spare relationship.**

```
mysql> select SupplierID from SUPPLIER_SPARE;
+------------+
| SupplierID |
+------------+
|       1201 |
|       1202 |
|       1203 |
|       1202 |
+------------+
4 rows in set (0.00 sec)

mysql> DELETE FROM SUPPLIER_SPARE WHERE SupplierID = 1203 AND SparePartID = 1103;
Query OK, 1 row affected (0.01 sec)

mysql> select SupplierID from SUPPLIER_SPARE;
+------------+
| SupplierID |
+------------+
|       1201 |
|       1202 |
|       1202 |
+------------+
3 rows in set (0.00 sec)
```

PART 4: BASIC SELECT QUERIES

1. **Display all assets**

```
mysql> SELECT *FROM ASSET;
+---------+----------------------+-----------------+-------------+---------+-------------+------------------+--------+-------------+
| AssetID | AssetName            | IdentificationNo | Model       | MnfYear | Status      | CriticalityLevel | UnitID | AssetTypeID |
+---------+----------------------+-----------------+-------------+---------+-------------+------------------+--------+-------------+
|     401 | Fighter Asset A-101  | IAF-SYN-101     | Fighter-Mk1 |    2023 | Degraded    | High             |    201 |         301 |
|     402 | Fighter Asset A-102  | IAF-SYN-102     | Fighter-Mk1 |    2022 | Degraded    | High             |    201 |         301 |
|     403 | Transport Asset T-201 | IAF-SYN-201     | Transport-X |    2021 | Ready       | Medium           |    202 |         302 |
|     404 | Naval Asset N-301    | IN-SYN-301      | Naval-X     |    2024 | Ready       | High             |    203 |         303 |
|     405 | Armoured Asset L-401 | IA-SYN-401      | Armour-X    |    2022 | Unavailable | High             |    204 |         304 |
+---------+----------------------+-----------------+-------------+---------+-------------+------------------+--------+-------------+
5 rows in set (0.00 sec)
```

2. **Display only unavailable assets**

```
mysql> SELECT AssetID, AssetName, Status FROM ASSET WHERE Status = 'Unavailable';
+---------+----------------------+-------------+
| AssetID | AssetName            | Status      |
+---------+----------------------+-------------+
|     405 | Armoured Asset L-401 | Unavailable |
+---------+----------------------+-------------+
1 row in set (0.01 sec)
```

3. **Display high-criticality assets**

```
mysql> SELECT AssetID, AssetName, CriticalityLevel FROM ASSET WHERE CriticalityLevel = 'High';
+---------+----------------------+------------------+
| AssetID | AssetName            | CriticalityLevel |
+---------+----------------------+------------------+
|     401 | Fighter Asset A-101  | High             |
|     402 | Fighter Asset A-102  | High             |
|     404 | Naval Asset N-301    | High             |
|     405 | Armoured Asset L-401 | High             |
+---------+----------------------+------------------+
4 rows in set (0.00 sec)
```

4. **Display assets ordered by manufacturing year**

```
mysql> SELECT AssetName, MnfYear FROM ASSET ORDER BY MnfYear DESC;
+----------------------+---------+
| AssetName            | MnfYear |
+----------------------+---------+
| Naval Asset N-301    |    2024 |
| Fighter Asset A-101  |    2023 |
| Fighter Asset A-102  |    2022 |
| Armoured Asset L-401 |    2022 |
| Transport Asset T-201 |    2021 |
+----------------------+---------+
5 rows in set (0.00 sec)
```

5. **Display different asset statuses**

```
mysql> SELECT DISTINCT Status FROM ASSET;
+-------------+
| Status      |
+-------------+
| Degraded    |
| Ready       |
| Unavailable |
+-------------+
3 rows in set (0.01 sec)
```

PART 5: AGGREGATE FUNCTIONS

1. **Count total assets**

```
mysql> SELECT COUNT(*) AS TotalAssets FROM ASSET;
+-------------+
| TotalAssets |
+-------------+
|           5 |
+-------------+
1 row in set (0.01 sec)
```

2. **Average maintenance cost**

```
mysql> SELECT AVG(CostIncurred) AS AverageMaintenanceCost FROM MAINTENANCE_EVENT;
+------------------------+
| AverageMaintenanceCost |
+------------------------+
|           55000.000000 |
+------------------------+
1 row in set (0.00 sec)
```

3. **Total maintenance cost**

```
mysql> SELECT SUM(CostIncurred) AS TotalMaintenanceCost FROM MAINTENANCE_EVENT;
+----------------------+
| TotalMaintenanceCost |
+----------------------+
|            110000.00 |
+----------------------+
1 row in set (0.00 sec)
```

4. **Maximum downtime**

```
mysql> SELECT MAX(DowntimeHours) AS MaximumDowntime FROM MAINTENANCE_EVENT;
+-----------------+
| MaximumDowntime |
+-----------------+
|            6.50 |
+-----------------+
1 row in set (0.00 sec)
```

5. **Minimum maintenance cost**

```
mysql> SELECT MIN(CostIncurred) AS MinimumMaintenanceCost FROM MAINTENANCE_EVENT;
+------------------------+
| MinimumMaintenanceCost |
+------------------------+
|               25000.00 |
+------------------------+
1 row in set (0.02 sec)
```

PART 6: GROUP BY

1. **Number of assets by status**

```
mysql> SELECT Status, COUNT(*) AS AssetCount FROM ASSET GROUP BY Status;
+-------------+------------+
| Status      | AssetCount |
+-------------+------------+
| Degraded    |          2 |
| Ready       |          2 |
| Unavailable |          1 |
+-------------+------------+
3 rows in set (0.01 sec)
```

2. **Number of assets by criticality**

```
mysql> SELECT CriticalityLevel, COUNT(*) AS AssetCount FROM ASSET GROUP BY CriticalityLevel;
+------------------+------------+
| CriticalityLevel | AssetCount |
+------------------+------------+
| High             |          4 |
| Medium           |          1 |
+------------------+------------+
2 rows in set (0.00 sec)
```

3. **Total maintenance cost by maintenance type**

```
mysql> SELECT MaintenanceType,SUM(CostIncurred) AS TotalCost FROM MAINTENANCE_EVENT GROUP BY MaintenanceType;
+-----------------+-----------+
| MaintenanceType | TotalCost |
+-----------------+-----------+
| Inspection      |  25000.00 |
| Repair          |  85000.00 |
+-----------------+-----------+
2 rows in set (0.01 sec)
```

PART 7: HAVING

1. **Asset statuses having more than one asset**

```
mysql> SELECT Status, COUNT(*) AS AssetCount FROM ASSET GROUP BY Status HAVING COUNT(*) > 1;
+----------+------------+
| Status   | AssetCount |
+----------+------------+
| Degraded |          2 |
| Ready    |          2 |
+----------+------------+
2 rows in set (0.00 sec)
```

2. **Maintenance types with total cost greater than 50000**

```
mysql> SELECT MaintenanceType,SUM(CostIncurred) AS TotalCost FROM MAINTENANCE_EVENT GROUP BY MaintenanceType HAVING SUM(CostIncurred) > 50000;
+-----------------+-----------+
| MaintenanceType | TotalCost |
+-----------------+-----------+
| Repair          |  85000.00 |
+-----------------+-----------+
1 row in set (0.01 sec)
```

PART 8: JOINS

1. Asset + Unit

```
mysql> SELECT a.AssetName,u.UnitName,u.Location FROM ASSET a JOIN UNIT u ON a.UnitID = u.UnitID;
+-----------------------+---------------------------+-------------+
| AssetName             | UnitName                  | Location    |
+-----------------------+---------------------------+-------------+
| Fighter Asset A-101   | Fighter Squadron Alpha    | Bengaluru   |
| Fighter Asset A-102   | Fighter Squadron Alpha    | Bengaluru   |
| Transport Asset T-201 | Transport Squadron Bravo  | Chandigarh  |
| Naval Asset N-301     | Naval Support Unit        | Kochi       |
| Armoured Asset L-401  | Armoured Unit Charlie     | Jaisalmer   |
+-----------------------+---------------------------+-------------+
5 rows in set (0.02 sec)
```

2. Asset + Unit + Service

```
mysql> SELECT a.AssetName,u.UnitName,s.ServiceName FROM ASSET a JOIN UNIT u ON a.UnitID =u.UnitID JOIN SERVICE s ON u.ServiceID= s.ServiceID;
+-----------------------+--------------------------+-----------------+
| AssetName             | UnitName                 | ServiceName     |
+-----------------------+--------------------------+-----------------+
| Fighter Asset A-101   | Fighter Squadron Alpha   | Indian Air Force |
| Fighter Asset A-102   | Fighter Squadron Alpha   | Indian Air Force |
| Transport Asset T-201 | Transport Squadron Bravo | Indian Air Force |
| Naval Asset N-301     | Naval Support Unit       | Indian Navy      |
| Armoured Asset L-401  | Armoured Unit Charlie    | Indian Army      |
+-----------------------+--------------------------+-----------------+
5 rows in set (0.01 sec)
```

3. Asset + Subsystem

```
mysql> SELECT a.AssetName,ss.SubsystemName,ss.CriticalityLevel FROM ASSET a JOIN SUBSYSTEM ss ON a.AssetID = ss.AssetID;
+-----------------------+-----------------+------------------+
| AssetName             | SubsystemName   | CriticalityLevel |
+-----------------------+-----------------+------------------+
| Fighter Asset A-101   | Engine System   | High             |
| Fighter Asset A-102   | Cooling System  | High             |
| Armoured Asset L-401  | Hydraulic System | High            |
| Fighter Asset A-101   | Power System    | High             |
+-----------------------+-----------------+------------------+
4 rows in set (0.01 sec)
```

4. Subsystem + Component

```
mysql> SELECT ss.SubsystemName,c.ComponentName,c.CriticalityLevel FROM SUBSYSTEM ss JOIN COMPONENT c ON ss.SubsystemID = c.SubsystemID;
+-----------------+--------------------+------------------+
| SubsystemName   | ComponentName      | CriticalityLevel |
+-----------------+--------------------+------------------+
| Engine System   | Fuel Pump          | High             |
| Engine System   | Engine Control Unit | High            |
| Cooling System  | Cooling Pump       | High             |
| Hydraulic System | Hydraulic Valve   | High             |
| Power System    | Power Controller   | High             |
+-----------------+--------------------+------------------+
5 rows in set (0.00 sec)
```

5. Component + Spare Part

```
mysql> SELECT c.ComponentName,sp.PartName,cs.QuantityRequired,cs.IsCritical
    -> FROM COMPONENT c JOIN COMPONENT_SPARE cs ON c.ComponentID = cs.ComponentID JOIN SPARE_PART sp ON cs.SparePartID = sp.SparePartID;
+-----------------+-------------------------+------------------+------------+
| ComponentName   | PartName                | QuantityRequired | IsCritical |
+-----------------+-------------------------+------------------+------------+
| Fuel Pump       | Fuel Pump Seal          |                2 | Yes        |
| Cooling Pump    | Cooling Pump Assembly   |                1 | Yes        |
| Hydraulic Valve | Hydraulic Valve Kit     |                1 | Yes        |
| Power Controller | Power Controller Module |               1 | Yes        |
+-----------------+-------------------------+------------------+------------+
4 rows in set (0.01 sec)
```

6. Spare Part + Supplier

```
mysql> SELECT sp.PartName,su.SupplierName,ss.UnitPrice,ss.LeadTimeDay,ss.PreferredSupplier
    -> FROM SPARE_PART sp JOIN SUPPLIER_SPARE ss ON sp.SparePartID = ss.SparePartID JOIN SUPPLIER su ON ss.SupplierID = su.SupplierID;
+-------------------------+---------------------------+-----------+-------------+-------------------+
| PartName                | SupplierName              | UnitPrice | LeadTimeDay | PreferredSupplier |
+-------------------------+---------------------------+-----------+-------------+-------------------+
| Fuel Pump Seal          | Bharat Defence Components |   5000.00 |           7 | Yes               |
| Cooling Pump Assembly   | Aero Systems Manufacturing |  85000.00 |          14 | Yes               |
| Power Controller Module | Aero Systems Manufacturing | 120000.00 |          21 | Yes               |
+-------------------------+---------------------------+-----------+-------------+-------------------+
3 rows in set (0.00 sec)
```

7. **Maintenance + User**

```
mysql> SELECT me.MaintenanceID,me.MaintenanceType,u.FullName AS PerformedBy,me.CostIncurred
    -> FROM MAINTENANCE_EVENT me JOIN `USER` u ON me.PerformedBy = u.UserID;
+---------------+-----------------+-------------+--------------+
| MaintenanceID | MaintenanceType | PerformedBy | CostIncurred |
+---------------+-----------------+-------------+--------------+
|           801 | Inspection      | Vikram Nair |     25000.00 |
|           802 | Repair          | Vikram Nair |     85000.00 |
+---------------+-----------------+-------------+--------------+
2 rows in set (0.01 sec)
```

PART 9: MULTIPLE-TABLE JOIN

1. **Show complete asset hierarchy**

```
mysql> SELECT s.ServiceName,u.UnitName,a.AssetName,ss.SubsystemName,c.ComponentName
    -> FROM SERVICE s
    -> JOIN UNIT u ON s.ServiceID = u.ServiceID
    -> JOIN ASSET a ON u.UnitID = a.UnitID
    -> JOIN SUBSYSTEM ss ON a.AssetID = ss.AssetID
    -> JOIN COMPONENT c ON ss.SubsystemID = c.SubsystemID;
+-----------------+-----------------------+---------------------+-----------------+--------------------+
| ServiceName     | UnitName              | AssetName           | SubsystemName   | ComponentName      |
+-----------------+-----------------------+---------------------+-----------------+--------------------+
| Indian Air Force | Fighter Squadron Alpha | Fighter Asset A-101 | Engine System   | Fuel Pump          |
| Indian Air Force | Fighter Squadron Alpha | Fighter Asset A-101 | Engine System   | Engine Control Unit |
| Indian Air Force | Fighter Squadron Alpha | Fighter Asset A-101 | Power System    | Power Controller   |
| Indian Air Force | Fighter Squadron Alpha | Fighter Asset A-102 | Cooling System  | Cooling Pump       |
| Indian Army     | Armoured Unit Charlie  | Armoured Asset L-401 | Hydraulic System | Hydraulic Valve    |
+-----------------+-----------------------+---------------------+-----------------+--------------------+
5 rows in set (0.00 sec)
```

PART 10: SUBQUERIES

1. **Assets having above-average maintenance cost**

```
mysql> SELECT DISTINCT a.AssetID, a.AssetName
    -> FROM ASSET a
    -> JOIN MAINTENANCE_WORK_ORDER wo
    ->     ON a.AssetID = wo.AssetID
    -> JOIN MAINTENANCE_EVENT me
    ->     ON wo.WorkOrderID = me.WorkOrderID
    -> WHERE me.CostIncurred > (
    ->     SELECT AVG(CostIncurred)
    ->     FROM MAINTENANCE_EVENT
    -> );
+---------+---------------------+
| AssetID | AssetName           |
+---------+---------------------+
|     402 | Fighter Asset A-102 |
+---------+---------------------+
1 row in set (0.02 sec)
```

2. **Assets with at least one failure**

```
mysql> SELECT AssetID, AssetName FROM ASSET WHERE AssetID IN(SELECT AssetID FROM FAILURE_EVENT);
+---------+----------------------+
| AssetID | AssetName            |
+---------+----------------------+
|     401 | Fighter Asset A-101  |
|     402 | Fighter Asset A-102  |
|     405 | Armoured Asset L-401 |
+---------+----------------------+
3 rows in set (0.01 sec)
```

3. **Spare parts below reorder level**

```
mysql> SELECT SparePartID, PartName,
    ->        StockQuantity, ReorderLevel
    -> FROM SPARE_PART
    -> WHERE StockQuantity <= ReorderLevel;
Empty set (0.00 sec)
```

PART 11: CORRELATED SUBQUERY

1. **Find spare parts whose stock is below their own reorder level.**

```
mysql> SELECT sp1.SparePartID,
    ->        sp1.PartName,
    ->        sp1.StockQuantity,
    ->        sp1.ReorderLevel
    -> FROM SPARE_PART sp1
    -> WHERE sp1.StockQuantity <= (
    ->     SELECT sp2.ReorderLevel
    ->     FROM SPARE_PART sp2
    ->     WHERE sp2.SparePartID = sp1.SparePartID
    -> );
Empty set (0.00 sec)
```

PART 12: SET OPERATIONS

1. **Assets that are Ready or Degraded**

```
mysql> SELECT AssetName
    -> FROM ASSET
    -> WHERE Status = 'Ready'
    ->
    -> UNION
    ->
    -> SELECT AssetName
    -> FROM ASSET
    -> WHERE Status = 'Degraded';
+------------------------+
| AssetName              |
+------------------------+
| Transport Asset T-201  |
| Naval Asset N-301      |
| Fighter Asset A-101    |
| Fighter Asset A-102    |
+------------------------+
4 rows in set (0.01 sec)
```

PART 13: VIEW

1. **View for asset readiness information.**

```
mysql> CREATE OR REPLACE VIEW asset_readiness_view AS
    -> SELECT
    ->     a.AssetID,
    ->     a.AssetName,
    ->     a.Status,
    ->     a.CriticalityLevel,
    ->     u.UnitName,
    ->     s.ServiceName
    -> FROM ASSET a
    -> JOIN UNIT u
    ->     ON a.UnitID = u.UnitID
    -> JOIN SERVICE s
    ->     ON u.ServiceID = s.ServiceID;
Query OK, 0 rows affected (0.07 sec)

mysql> SELECT *
    -> FROM asset_readiness_view;
+---------+----------------------+-------------+-----------------+--------------------------+-----------------+
| AssetID | AssetName            | Status      | CriticalityLevel | UnitName                 | ServiceName     |
+---------+----------------------+-------------+-----------------+--------------------------+-----------------+
|     401 | Fighter Asset A-101  | Degraded    | High            | Fighter Squadron Alpha   | Indian Air Force |
|     402 | Fighter Asset A-102  | Degraded    | High            | Fighter Squadron Alpha   | Indian Air Force |
|     403 | Transport Asset T-201 | Ready      | Medium          | Transport Squadron Bravo | Indian Air Force |
|     404 | Naval Asset N-301    | Ready       | High            | Naval Support Unit       | Indian Navy      |
|     405 | Armoured Asset L-401 | Unavailable | High            | Armoured Unit Charlie    | Indian Army      |
+---------+----------------------+-------------+-----------------+--------------------------+-----------------+
5 rows in set (0.01 sec)
```

PART 14: INDEXES

1. **Frequently searched asset status**

```
mysql> CREATE INDEX idx_asset_status ON ASSET(Status);
Query OK, 0 rows affected (0.26 sec)
Records: 0  Duplicates: 0  Warnings: 0
```

2. **Frequently searched component criticality**

```
mysql> CREATE INDEX idx_component_criticality ON COMPONENT(CriticalityLevel);
Query OK, 0 rows affected (0.11 sec)
Records: 0  Duplicates: 0  Warnings: 0
```

3. **Failure lookup by asset**

```
mysql> CREATE INDEX idx_failure_asset ON FAILURE_EVENT(AssetID);
Query OK, 0 rows affected (0.09 sec)
Records: 0  Duplicates: 0  Warnings: 0
```

4. **Maintenance lookup by work order**

```
mysql> CREATE INDEX idx_maintenance_workorder ON MAINTENANCE_EVENT(WorkOrderID);
Query OK, 0 rows affected (0.09 sec)
Records: 0  Duplicates: 0  Warnings: 0
```

PART 15: TRANSACTION

1. **Example: Maintenance consumes spare parts. Both stock update and maintenance record should succeed together.**

```
mysql> START TRANSACTION;
Query OK, 0 rows affected (0.00 sec)

mysql>
mysql> UPDATE SPARE_PART
    -> SET StockQuantity = StockQuantity - 2
    -> WHERE SparePartID = 1101
    ->   AND StockQuantity >= 2;
Query OK, 1 row affected (0.02 sec)
Rows matched: 1  Changed: 1  Warnings: 0

mysql>
mysql> COMMIT;
Query OK, 0 rows affected (0.01 sec)
```

2. **If something goes wrong:**

```
mysql> ROLLBACK;
Query OK, 0 rows affected (0.00 sec)

mysql>
```

PART 16: TRIGGER

1. **Trigger that automatically updates an asset's status to `Ready` whenever a new maintenance event of type `Repair` is inserted**

```
mysql> CREATE TRIGGER trg_asset_repair_status
    -> AFTER INSERT ON maintenance_event
    -> FOR EACH ROW
    -> BEGIN
    ->     IF NEW.MaintenanceType = 'Repair' THEN
    ->         UPDATE asset
    ->         SET Status = 'Ready'
    ->         WHERE AssetID = (
    ->             SELECT AssetID
    ->             FROM maintenance_work_order
    ->             WHERE WorkOrderID = NEW.WorkOrderID
    ->         );
    ->     END IF;
    -> END$$
Query OK, 0 rows affected (0.04 sec)
```

Testing the trigger

```
mysql> SELECT AssetID, AssetName, Status
    -> FROM asset
    -> WHERE AssetID = 402;
+---------+---------------------+----------+
| AssetID | AssetName           | Status   |
+---------+---------------------+----------+
|     402 | Fighter Asset A-102 | Degraded |
+---------+---------------------+----------+
1 row in set (0.00 sec)

mysql> INSERT INTO maintenance_event
    ->     (MaintenanceID, WorkOrderID, PerformedBy,
    ->      MaintenanceType, Remarks)
    -> VALUES
    ->     (803, 701, 103, 'Repair',
    ->      'Trigger testing');
Query OK, 1 row affected (0.02 sec)

mysql> SELECT AssetID, AssetName, Status
    -> FROM asset
    -> WHERE AssetID = 402;
+---------+---------------------+--------+
| AssetID | AssetName           | Status |
+---------+---------------------+--------+
|     402 | Fighter Asset A-102 | Ready  |
+---------+---------------------+--------+
1 row in set (0.00 sec)
```

PART 17: STORED FUNCTION

1. **Returns the number of failures for an asset.**

```
mysql> DROP FUNCTION IF EXISTS get_failure_count;
Query OK, 0 rows affected, 1 warning (0.08 sec)

mysql>
mysql> DELIMITER $$
mysql>
mysql> CREATE FUNCTION get_failure_count(
    ->     p_asset_id INT
    -> )
    -> RETURNS INT
    -> DETERMINISTIC
    -> READS SQL DATA
    -> BEGIN
    ->     DECLARE failure_count INT DEFAULT 0;
    ->
    ->     SELECT COUNT(*)
    ->     INTO failure_count
    ->     FROM FAILURE_EVENT
    ->     WHERE AssetID = p_asset_id;
    ->
    ->     RETURN failure_count;
    -> END$$
Query OK, 0 rows affected (0.02 sec)

mysql>
mysql> DELIMITER ;
mysql>
```

2. **Call the function**

```
mysql> SELECT get_failure_count(401);
+------------------------+
| get_failure_count(401) |
+------------------------+
|                      1 |
+------------------------+
1 row in set (0.01 sec)
```

PART 18: STORED PROCEDURE

1. **Updates the status of an asset.**

```
mysql> DROP PROCEDURE IF EXISTS update_asset_status;
Query OK, 0 rows affected, 1 warning (0.00 sec)

mysql>
mysql> DELIMITER $$
mysql>
mysql> CREATE PROCEDURE update_asset_status(
    ->     IN p_asset_id INT,
    ->     IN p_status VARCHAR(50)
    -> )
    -> BEGIN
    ->     UPDATE ASSET
    ->     SET Status = p_status
    ->     WHERE AssetID = p_asset_id;
    -> END$$
Query OK, 0 rows affected (0.02 sec)

mysql>
mysql> DELIMITER ;
mysql>
```

```
mysql> CALL update_asset_status(401, 'Ready');
Query OK, 1 row affected (0.01 sec)

mysql> SELECT AssetID, AssetName, Status
    -> FROM ASSET
    -> WHERE AssetID = 401;
+---------+---------------------+--------+
| AssetID | AssetName           | Status |
+---------+---------------------+--------+
|     401 | Fighter Asset A-101 | Ready  |
+---------+---------------------+--------+
1 row in set (0.00 sec)
```

PART 19: CTE

1. **Calculate total maintenance cost for each work order.**

```
mysql> WITH MaintenanceSummary AS (
    ->     SELECT
    ->         WorkOrderID,
    ->         SUM(CostIncurred) AS TotalCost
    ->     FROM MAINTENANCE_EVENT
    ->     GROUP BY WorkOrderID
    -> )
    -> SELECT *
    -> FROM MaintenanceSummary
    -> ORDER BY TotalCost DESC;
+-------------+-----------+
| WorkOrderID | TotalCost |
+-------------+-----------+
|         701 |  85000.00 |
|         702 |  25000.00 |
+-------------+-----------+
2 rows in set (0.00 sec)
```

PART 20: WINDOW FUNCTION

1. **Rank maintenance events by cost.**

```
mysql> SELECT
    ->     MaintenanceID,
    ->     MaintenanceType,
    ->     CostIncurred,
    ->     RANK() OVER (
    ->         ORDER BY CostIncurred DESC
    ->     ) AS CostRank
    -> FROM MAINTENANCE_EVENT;
+---------------+-----------------+--------------+----------+
| MaintenanceID | MaintenanceType | CostIncurred | CostRank |
+---------------+-----------------+--------------+----------+
|           802 | Repair          |     85000.00 |        1 |
|           801 | Inspection      |     25000.00 |        2 |
|           803 | Repair          |         NULL |        3 |
+---------------+-----------------+--------------+----------+
3 rows in set (0.00 sec)
```

PART 21: USEFUL PROJECT QUERY

1. **Find unavailable or degraded assets and their latest known failure.**

```
mysql> SELECT
    ->     a.AssetID,
    ->     a.AssetName,
    ->     a.Status,
    ->     f.FailureType,
    ->     f.Severity,
    ->     f.OccurredAt
    -> FROM ASSET a
    -> JOIN FAILURE_EVENT f
    ->     ON a.AssetID = f.AssetID
    -> WHERE a.Status IN ('Unavailable', 'Degraded')
    -> ORDER BY f.OccurredAt DESC;
+---------+---------------------+-------------+-----------------+----------+---------------------+
| AssetID | AssetName           | Status      | FailureType     | Severity | OccurredAt          |
+---------+---------------------+-------------+-----------------+----------+---------------------+
|     405 | Armoured Asset L-401 | Unavailable | Hydraulic Failure | Medium   | 2026-09-21 14:15:00 |
+---------+---------------------+-------------+-----------------+----------+---------------------+
1 row in set (0.00 sec)
```

2. **Find critical components and the spare parts they require.**

```
mysql> SELECT
    ->     c.ComponentName,
    ->     c.CriticalityLevel,
    ->     sp.PartName,
    ->     sp.StockQuantity,
    ->     sp.ReorderLevel
    -> FROM COMPONENT c
    -> JOIN COMPONENT_SPARE cs
    ->     ON c.ComponentID = cs.ComponentID
    -> JOIN SPARE_PART sp
    ->     ON cs.SparePartID = sp.SparePartID
    -> WHERE c.CriticalityLevel = 'High';
+-----------------+-----------------+-------------------------+--------------+--------------+
| ComponentName   | CriticalityLevel | PartName                | StockQuantity | ReorderLevel |
+-----------------+-----------------+-------------------------+--------------+--------------+
| Fuel Pump       | High            | Fuel Pump Seal          |           48 |           10 |
| Cooling Pump    | High            | Cooling Pump Assembly   |            8 |            3 |
| Hydraulic Valve | High            | Hydraulic Valve Kit     |           15 |            5 |
| Power Controller | High           | Power Controller Module |            6 |            2 |
+-----------------+-----------------+-------------------------+--------------+--------------+
4 rows in set (0.01 sec)
```

3. **Find components whose required spare parts are below reorder level.**

```
mysql> SELECT
    ->     c.ComponentName,
    ->     sp.PartName,
    ->     sp.StockQuantity,
    ->     sp.ReorderLevel
    -> FROM COMPONENT c
    -> JOIN COMPONENT_SPARE cs
    ->     ON c.ComponentID = cs.ComponentID
    -> JOIN SPARE_PART sp
    ->     ON cs.SparePartID = sp.SparePartID
    -> WHERE sp.StockQuantity <= sp.ReorderLevel;
Empty set (0.00 sec)
```

4. **Find component dependencies.**

```
mysql> SELECT
    ->     c1.ComponentName AS Component,
    ->     c2.ComponentName AS DependsOn
    -> FROM COMPONENT_DEPENDENCY cd
    -> JOIN COMPONENT c1
    ->     ON cd.ComponentID = c1.ComponentID
    -> JOIN COMPONENT c2
    ->     ON cd.DependsOnComponentID = c2.ComponentID;
+---------------------+-----------+
| Component           | DependsOn |
+---------------------+-----------+
| Engine Control Unit | Fuel Pump |
| Cooling Pump        | Fuel Pump |
+---------------------+-----------+
2 rows in set (0.01 sec)
```